NORMAL CHECKLIST

A340-300

BEFORE START

COCKPIT PREP.....COMPLETED (BOTH)
GEAR PINS and COVERS.....REMOVED
SIGNS.....ON / AUTO
ADIRS.....NAV
FUEL QUANTITY.....____ KG.LB
TO DATA.....____ SET
BARO REF.....____ SET (BOTH)

WINDOWS/DOORS.....CLOSED/ARMED (BOTH)
BEACON.....ON
THR LEVERS.....IDLE
PARKING BRAKE.....AS RQRD

AFTER START

ANTI-ICE.....AS RQRD
ECAM STATUS.....CHECKED
PITCH TRIM.....____ SET
RUDDER TRIM.....NEUTRAL

BEFORE TAKEOFF

FLT CTRL.....CHECKED (BOTH)
FLT INST.....CHECKED (BOTH)
BRIEFING.....CONFIRMED
FLAP SETTING.....CONF____ (BOTH)
V1.VR.V2/FLX TEMP.....____ (BOTH)
ATC.....SET
ECAM MEMO.....TO NO BLUE
- AUTO BRK MAX
- SIGNS ON
- SPLRS ARM
- FLAPS T.O
- T.O CONFIG NORMAL

CABIN CREW.....ADVISED
LIGHTS.....ON
ENG MODE SEL.....AS RQRD
PACKS.....ON APU/OFF
RADAR/TCAS/PWS.....AS RQRD/SET/AUTO

AFTER TAKEOFF / CLIMB

LDG GEAR.....UP
FLAPS.....RETRACTED
PACKS / APU.....ON/OFF

BARO REF.....____ SET (BOTH)

APPROACH

BRIEFING.....CONFIRMED
ECAM STATUS.....CHECKED
SEAT BELTS.....ON
MDA/DH.....____ (BOTH)
ENG MODE SEL.....AS RQRD

BARO REF.....____ (BOTH)

LANDING

CABIN CREW.....ADVISED
A/THR.....SPEED/OFF
ECAM MEMO.....LDG NO BLUE
- SIGNS ON
- CABIN CHECK
- LDG GEAR DN
- FLAPS LDG
- SPLRS ARM

AFTER LANDING

SPOILERS.....DISARMED
FLAPS.....RETRACTED
RADAR.....OFF/STBY
APU.....START

PARKING

APU BLEED.....ON
ENGINES.....OFF
SEAT BELTS.....OFF
EXT LT.....AS RQRD
FUEL PUMPS.....OFF
PARK BRK and CHOCKS.....AS RQRD

SECURING THE AIRCRAFT

TCAS.....STBY
ADIRS.....OFF
OXYGEN.....OFF
APU BLEED.....OFF
EMER EXIT LT.....OFF
NO SMOKING.....OFF
EXT PWR.....AS RQRD
BATT 1 + 2 + APU BAT.....AS RQRD
APU AND BAT.....OFF

EMERGENCY EVACUATION

AIRCRAFT/PARKING BRAKE.....STOP / ON
ATC (VHF 1).....NOTIFY
CABIN CREW (PA).....ALERT
ΔP (Only is MAN CAB PR has been used).....CHECK ZERO

- **If ΔP not at zero:**

  CAB PR MODE SEL.....MAN
  V/S CTL.....FULL UP

ALL ENG MASTERS.....OFF
ALL FIRE pb (ENGs & APU).....PUSH
ALL AGENTS (ENGs & APU).....AS RQRD

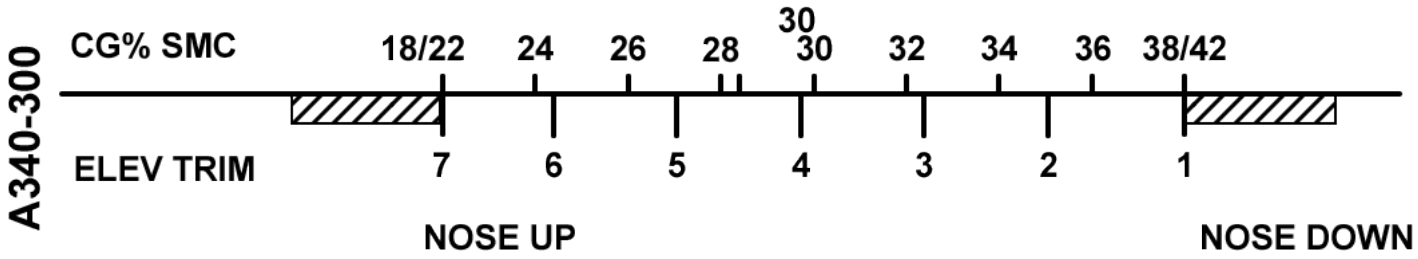
■ **If evacuation required:**

EVACUATION.....INITIATE
ALL BATs.....OFF

■ **If evacuation NOT required:**

CABIN CREW AND PASSENGERS (PA).....NOTIFY

TAKEOFF CG/TRIM



AUTOMATIC ENGINE START

ENG START selector.....**IGN/START**
The lower ECAM display shows ENG page

ANNOUNCE.....**"STARTING ENGINES 1 and 2"**
Engines 1 and 2 are usually started first (Engine 2 powers blue hydraulic system, which pressurizes the parking brake).

Do not turn MASTER switch ON before all amber crosses and messages have disappeared on the engine parameters (upper ECAM display).

MASTER switch 1 then 2.....**ON**
Wait until the ENG 1 N2 is at about 10 % before switching the ENG 2 MASTER switch at ON (to permit correct operation of the No Break Power Transfer).

Monitor start of both engines

ON ECAM UPPER DISPLAY	ON ECAM LOWER DISPLAY
N2 increases	Corresponding start valve in line Bleed pressure indication green Oil pressure increases
At 16 % N2:	Indication of active igniter (A or B) appears
At 22 % N2: – FF increases 10 seconds (maximum) after fuel is on: – EGT increases – N1 increases	
At 50 % N2:	Start valve crossline Igniter is turned OFF

ENG. IDLE PARAMETERS.....**CHECK NORMAL**
At ISA sea level:
N1 about 20%
N2 about 60%
EGT about 400°C
FF 300 kg/h (660 lb/h)

ANNOUNCE.....**"STARTING ENGINES 3 and 4"**

ENG MASTER 3 then 4 (ENG 4, when ENG 3's N2 is at about 10 %).....**ON**
Same procedure as for Engines 1 & 2

MANUAL ENGINE START

THR LEVERS.....**IDLE**

ENG MODE selector.....**NORM THEN IGN/START**
The lower ECAM displays engine page

NOTE: If all engines are started manually, the following procedure applies one engine at a time.

- **When all engines parameters are available on the upper ECAM display (no amber crosses displayed):**
  ENG MAN START pb-sw.....**ON**
  START VALVE.....**CHECK IN-LINE**
  OIL PRESSURE.....**CHECK**
  N2 INCREASE.....**CHECK**

- **When N2 reaches maximum motoring speed (minimum 20 %):**

  MASTER Switch.....**ON**
  CHRONO.....**START**
  The PM starts the timing in order to monitor the light-up delay
  IGNITERS A AND B.....**CHECK ON**
  FUEL FLOW INCREASE.....**CHECK**
  EGT INCREASE.....**CHECK**
  N1 INCREASE.....**CHECK**

- **When idle is reached (AVAIL indication is displayed):**
  START VALVE.....**CHECK CROSS LINE**
  IGNITERS A AND B.....**CHECK OFF**
  MAIN ENG PARAMETERS.....**CHECK NORMAL**
  SECONDARY ENG PARAMETERS.....**CHECK NORMAL**
  ENG MAN START pb-sw.....**OFF**
  ENG START selector.....**NORM**

LOSS OF BRAKING

- **If Autobrake is selected:**

BRAKE PEDALS.....**PRESS**

- **If no braking:**

REV.....**MAX**

If needed, the flight crew may keep maximum reverse thrust until full aircraft stop

BRAKE PEDALS.....**RELEASE**

A/SKID & N/W STRG.....**OFF**
BRAKE PEDALS.....**PRESS**
MAX BRK PR.....**1000 PSI**

Limit brake pressure to approximately 1 000 PSI. Monitor brake pressure on the BRAKES PRESS indicator

- **If still no braking:**

PARK BRAKE.....**USE**

Use short successive parking brake applications to stop the aircraft.

EMER DESCENT

CREW OXY MASKS.....**USE**
SIGNS.....**ON**
EMER DESCENT.....**INITIATE**

- **If A/THR not active:**

THR LEVERS.....**IDLE**

SPEED BRAKES.....**FULL**

- **When descent established:**

SPEED.....**MAX / APPROPRIATE**

MAX FL.....**100/MEA-MORA**

- **If structural damage suspected:**

MANEUVER WITH CARE

CONSIDER L/G EXTENSION

SIGNS.....**ON**

ENG START selector.....**IGN**

ATC.....**NOTIFY**

EMER DESCENT (PA).....**ANNOUNCE**

ATC XPDR 7700.....**CONSIDER**

- **If CAB ALT above 14 000 ft:**
  OXYGEN PAX MASK MAN ON.....**PRESS**

STALL RECOVERY

NOSE DOWN PITCH CONTROL.....**APPLY**

In case of lack of pitch down authority, reducing thrust may be necessary

BANK.....**WINGS LEVEL**

- **When out of stall (no longer stall indications):**
  THRUST.....**INCREASE SMOOTHLY AS NEEDED**

  SPEED BRAKES.....**CHECK RETRACTED**

  FLIGHT PATH.....**RECOVER SMOOTHLY**

- **If in clean configuration and below 20 000 ft:**
  FLAP 1.....**SELECT**

STALL WARNING AT LIFTOFF

THRUST.....**TOGA**

At the same time:

PITCH ATTITUDE.....**12.5 °**

BANK.....**WINGS LEVEL**

GPWS/EGPWS Warnings

- **“PULL UP” – “TERRAIN AHEAD PULL UP” – “OBSTACLE AHEAD PULL UP” – “AVOID TERRAIN” – “AVOID OBSTACLE”**

  Simultaneously:

AP	**OFF**
PITCH	**PULL UP**

  Pull to full backstick and maintain in that position

THRUST LEVERS	**TOGA**
SPEED BRAKES lever	**CHECK RETRACTED**
BANK	**WINGS LEVEL or ADJUST**

  DO NOT CHANGE CONFIGURATION (SLATS/FLAPS, GEAR) UNTIL CLEAR OF OBSTACLE.

TCAS WARNING – RESOLUTION ADVISORY

Always follow RA orders, even if this results in crossing the intruder altitude, because these orders ensure the best altitude separation.

CAUTION	Be aware that the intruder may have a TCAS, and may maneuver in response to a coordinated RA order. Therefore, not following an RA order could compromise safe separation.

■ **If the AP/FD TCAS mode is available:**

The flight crew applies this procedure, when a RA triggered, and the AP/FD TCAS mode is engaged. The AP/FD TCAS mode follows the RA orders.

- **All RA, except any CLIMB RA during approach in CONF 3 or FULL:**
  - **If the AP is OFF:**

FD ORDERs.....**FOLLOW**

The AP can be engaged

VERTICAL SPEED.....**MONITOR**

If a preventative RA was triggered: Check that the vertical speed remains out of the red areas of the vertical speed scale.

If a corrective RA was triggered: Check that the vertical speed remains out of the red area, and remains in the green area of the vertical speed scale.

CAUTION	If for any reason during a RA, the aircraft vertical speed does not reach the green area of the vertical speed scale, the Pilot Flying (PF) should disconnect the AP, and override the FD orders, in order to lead the aircraft vertical speed out of the red area of the vertical speed scale. If necessary, the PF must use the full speed range between Vαmax and Vmax.

- **Any CLIMB RA during approach in CONF 3 or FULL:**

GO-AROUND.....**PERFORM**

The AP/FD TCAS mode disengages (the AP/FD no longer follows the RA orders).
Follow the SRS GA mode.

VERTICAL SPEED.....**MONITOR**

Check that the vertical speed remains out of the red area of the vertical speed scale and take over if necessary.

Respect stall, GPWS or windshear warnings.

ATC.....**NOTIFY**

- **When the "CLEAR OF CONFLICT" aural alert sounds:**

The AP/FD TCAS mode, if engaged, disengages

AP/FD	**MONITOR/FOLLOW**
ATC	**NOTIFY**
LATERAL AND VERTICAL GUIDANCE	**ADJUST**

The flight crew should engage and appropriate vertical mode, or adjust the vertical speed target, in accordance with the latest ATC clearance.

SPEED	**ADJUST**

The flight crew should adjust the speed target, and revert to managed speed, as appropriate.

- **If the AP/FD TCAS mode is not available:**

The flight crew must apply this procedure, when a RA is triggered and the AP/FD TCAS mode does not engage.

- **All RA, except CLIMB RA during approach in CONF 3 or FULL:**

AP (if engaged)	**OFF**
BOTH FDs	**OFF**

Respond promptly and smoothly.

VERTICAL SPEED	**ADJUST or MAINTAIN**

Adjusting or maintain the vertical speed as required, to reach the green area and/or avoid the red area of the vertical speed scale.

Note: Avoid excessive maneuvers while attempting to maintain the vertical speed just outside the red area of the vertical speed scale, and within the green area. If necessary, use the full speed range between Vαmax and Vmax.

- **Any CLIMB RA during approach in CONF 3 or FULL:**

GO-AROUND	**PERFORM**

Follow the SRS GA mode.

VERTICAL SPEED	**MONITOR**

Check that the vertical speed remains out of the red area of the vertical speed scale and take over if necessary.

Respect stall, GPWS or windshear warnings.

ATC	**NOTIFY**

- **When the "CLEAR OF CONFLICT" aural alert sounds:**

ATC	**NOTIFY**
LATERAL AND VERTICAL GUIDANCE	**ADJUST**

Adjust the lateral and vertical guidance to resume normal navigation, in accordance with ATC clearance.

AP/FD	**AS RQRD**

If necessary, reengage the AP/FD.

WINDSHEAR WARNING – REACTIVE WINDSHEAR

If windshear is detected, both PFDs display a WINDSHEAR message. An associated aural WINDSHEAR alert sounds three times.

If the flight crew detects WINDSHEAR, apply the following recovery technique:

- **Takeoff:**
  - **If before V1:**

    If there are significant variations in airspeed, and in airspeed trend below the indicated V_1, reject the takeoff.

    If windshear occurs during the takeoff roll, the aircraft may reach V1 later (or sooner) than expected. The flight crew should ensure that there is sufficient runway remaining to stop the aircraft, if necessary.

  - **If after V1:**

THR LEVERS	**TOGA**
REACHING VR	**ROTATE**
SRS ORDERS	**FOLLOW**

If necessary, the flight crew may pull the sidestick fully back.

NOTE: If the FD bars are not displayed, move toward an initial pitch attitude of 17.5 °. Then, if necessary, to prevent a loss in altitude, increase the pitch attitude.

- **Airborne, initial climb or landing:**

THR LEVERS AT TOGA	**SET OR CONFIRM**
AP (if engaged)	**KEEP ON**
SRS ORDERS	**FOLLOW**

If necessary the flight crew may pull the sidestick fully back.

NOTE: 1. Autopilot disengages if the angle of attack value goes above alpha prot.
2. If the FD bars are not displayed, move toward an initial pitch attitude of 17.5 °. Then, if necessary, to prevent a loss in altitude, increase the pitch attitude.

DO NOT CHANGE CONFIGURATION (SLATS/FLAPS, GEAR) UNTIL OUT OF WINDSHEAR. CLOSELY MONITOR FLIGHT PATH AND SPEED. RECOVER SMOOTHLY TO NORMAL CLIMB OUT OF WINDSHEAR.